

streamguard-ai-793289044855.us-central1.run.app/dev-ui/?app=streamguard_agent&userId=user&session=368b9566-526c-48dd-b54d-00c0df41301a

Agent Development Kit 2.8.0streamguard_agentFocus only on Grafana...New Session

InfoStateArtifactsEvalsEventsTraces+ Filter

Event 108 of 108

Invocation: #4 (Investigate whether streaming degra...)



Shared-Origin Resource Starvation: It is hypothesized that because all regional streams share a single transcoding instance, they compete for CPU resources. Since video chunk/segment writing is scheduled at slightly offset regional boundaries, the transcoding segments clash and momentarily overload the central processor, triggering cascading latency and buffering.

Kafka Publisher Failure: It is hypothesized that the empty incident topics are due to either:

- A very short Kafka retention configuration that quickly pruned active degradation records.
- The shared machine's local stream-reporting pipeline crashing during the initial CPU overload at 09:18:25 UTC, preventing the host from logging subsequent incidents or alert state changes to the Confluent Kafka cluster.

#107

Investigate whether streaming degradation is affecting specific regions. Use Grafana telemetry and Confluent Kafka events where regional information is available. Identify the affected regions using only verified telemetry. If a Kafka event does not contain a region or region_id, explicitly state that regional correlation is not possible for that event.

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An investigation into whether streaming degradation is affecting specific regions has been performed by cross-referencing verified telemetry from **Grafana** (Prometheus metrics, Loki logs, and annotations) and events retrieved from **Confluent Kafka**.

1. Regional Analysis via Grafana Telemetry

Grafana's Prometheus metrics and Loki logs directly verify that all four monitored regions (Midwest, Northeast, South, and West) are actively experiencing severe streaming degradation.

The degradation is characterized by synchronized, recurring performance collapses (occurring on a roughly 60-second cycle) due to transcode latency and CPU spikes on the shared origin instance.

Below is the verified telemetry for each region:

A. Midwest Region (Verified Affected)

- Buffering & Video stuttering:** Buffering ratios spike periodically to 5.07% – 7.62% (normal baseline is <0.8%).
- CPU Starvation:** Shared host CPU usage spikes to 88.02% – 96.89% during degradation windows.
- Dropped Frames:** Dropped frames surge up to 454 to 487 frames per minute (normal is <10\$).

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